

RENMARK, Australia

WMO: 946870

Lat: 34.167S

Lon: 140.750E

Elev: 21

StdP: 101.07

Time Zone: 9.50 (AUA)

Period: 90-01

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.4	4.5	-1.7	3.3	20.1	-0.2	3.7	17.3	11.8	12.5	9.0	12.7	0.2	140	0.364

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.6	38.9	20.6	36.7	20.3	34.7	19.6	23.5	30.7	22.3	30.9	21.3	31.1	2.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.6	16.3	26.5	19.6	14.4	26.1	18.0	13.0	25.4	70.1	29.5	65.5	30.7	61.7	31.4	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.2	5.5	0.5	42.7	1.8	1.7	-0.8	43.9	-1.9	44.9	-2.9	45.9	-4.2	47.1		
(5)			0.5	25.3	1.1	1.5	-0.3	26.4	-1.0	27.2	-1.6	28.1	-2.3	29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.8	24.6	25.3	21.1	17.1	14.0	12.0	11.2	12.0	15.0	17.8	20.4	23.0		
	DBStd	5.88	4.07	4.06	3.85	3.06	2.57	2.27	1.95	2.45	3.29	3.70	4.14	3.89		
	HDD10.0	23	0	0	0	0	1	6	9	7	1	0	0	0		
	HDD18.3	1012	1	1	12	61	137	191	223	196	111	56	21	4		
	CDD10.0	2853	452	428	345	213	126	65	45	71	151	242	313	403		
	CDD18.3	799	195	196	98	23	3	1	0	1	11	39	84	148		
	CDH23.3	9421	2394	2363	1033	252	19	2	0	11	128	478	983	1758		
	CDH26.7	4603	1268	1272	468	71	1	0	0	1	32	191	442	858		
(14) Wind	WSAvg	1.7	1.9	1.6	1.6	1.2	1.1	1.2	1.5	1.8	1.9	2.2	1.9	2.1		
(15) Precipitation	PrecAvg	259	18	18	12	19	22	21	27	25	29	30	22	17		
	PrecMax	517	97	103	81	131	91	61	70	75	113	108	108	68		
	PrecMin	90	0	0	0	0	0	1	2	1	3	0	0	0		
	PrecStd	88	19	23	16	24	17	15	16	14	24	28	22	16		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.9	41.1	38.0	31.2	25.8	22.2	21.2	24.8	30.0	35.7	38.2	39.3		
		MCWB	22.1	22.1	20.1	17.3	15.2	13.9	12.4	13.9	15.7	17.5	20.0	20.0		
	2%	DB	38.8	38.7	34.7	28.9	23.4	19.1	18.6	21.7	26.6	31.5	35.0	36.5		
		MCWB	20.7	21.2	19.6	17.2	15.2	13.0	11.8	12.6	15.0	16.4	18.8	19.0		
	5%	DB	36.2	36.1	31.9	26.5	21.3	17.4	16.9	19.5	24.0	28.2	31.6	34.1		
		MCWB	20.6	20.8	18.9	16.7	14.7	13.0	11.3	11.9	14.6	15.9	17.5	18.7		
	10%	DB	33.7	33.8	29.3	23.9	19.4	16.2	15.5	17.7	21.5	25.6	28.8	31.5		
		MCWB	19.8	19.9	18.3	16.0	13.8	12.2	10.9	11.3	13.6	15.4	17.1	18.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.5	25.1	22.3	20.2	17.3	15.7	13.9	15.4	18.1	19.5	21.5	22.5		
		MCDB	34.3	34.5	32.4	26.8	21.6	18.2	18.1	21.7	24.2	29.1	32.1	29.8		
	2%	WB	23.4	23.9	20.9	18.6	16.4	14.5	12.7	13.9	16.5	18.0	20.2	21.3		
		MCDB	31.5	30.6	30.7	24.6	21.0	17.9	16.4	19.7	23.8	26.7	31.0	29.5		
	5%	WB	22.3	22.8	19.8	17.3	15.2	13.6	11.9	12.6	15.3	17.1	19.2	20.3		
		MCDB	31.5	29.6	29.8	24.3	19.9	16.8	15.7	18.1	22.2	24.9	28.7	29.4		
	10%	WB	21.1	21.4	18.9	16.3	14.3	12.6	11.2	11.6	14.1	16.0	18.1	19.1		
		MCDB	31.1	31.5	28.3	23.4	18.7	15.6	14.9	16.8	21.0	23.8	27.0	29.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.0	13.6	12.6	12.4	10.2	8.5	9.2	10.6	11.4	12.7	12.9	13.7		
		MCDBR	17.1	16.2	16.1	16.1	12.8	10.1	11.7	14.3	15.1	17.2	16.8	17.4		
	5% WB	MCWBR	6.0	5.1	5.7	7.1	6.5	5.7	6.1	6.9	6.1	6.4	6.3	5.7		
		MCWBR	13.0	11.7	13.7	14.1	10.4	8.9	10.0	12.2	13.4	13.6	14.4	12.9		
(40) Clear-Sky Solar Irradiance	taub	0.342	0.338	0.321	0.313	0.300	0.293	0.292	0.303	0.334	0.334	0.344	0.344	0.335		
	taud	2.498	2.533	2.567	2.563	2.583	2.590	2.589	2.543	2.445	2.462	2.467	2.467	2.507		
	Ebn at Noon	995	978	960	912	871	852	872	909	929	971	987	1006	1006		
	Edh at Noon	114	106	96	85	74	68	72	85	106	113	117	117	114		
(44) All-Sky Solar Radiation	RadAvg	7.92	7.09	5.87	4.30	3.03	2.48	2.68	3.64	4.94	6.32	7.21	7.88			
	RadStd	0.39	0.39	0.31	0.22	0.14	0.14	0.12	0.23	0.34	0.38	0.34	0.29			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+107	+69

Nomenclature: See separate page